

Hydra Methology

Introduction

The original purpose of the Hydra simulation model was to create simulated data for testing simpler multi-species assessment models (Gaichas et al. 2017). Further model development repurposed Hydra as an MSE tool to test proposed multispecies and ecosystem level management procedures to support EBFM analyses for the New England Fishery Management Council, and were reviewed in 2018 as part of an [Ecosystem Based Management Strategy Review for Georges Bank](#).

The current Hydra estimation model can still be used for the above purposes (no simulation code was removed), but with the potential added benefit of estimating operating model parameters by fitting to data from the ecosystem, rather than iterative hand calibration.

Multispecies models “of intermediate complexity” between single population and full ecosystem models potentially combine the best aspects of current single species assessment models (including demographic structure and statistical parameter estimation) with key ecological linkages between species (Plagányi et al. 2012, Collie et al. 2014). The Northeast Fisheries Science Center (NEFSC) is currently applying multispecies production and delay-difference models to a simulated system coarsely modeled after the Georges Bank fish community. The focus of this work is to evaluate the performance of a length-based multispecies approach for providing management advice. While we do not yet have a fitted model ready to provide catch advice, this review should identify next steps for further developing model code, inputs, and diagnostics towards this goal.

To evaluate Hydra’s performance as an estimation model, we converted it from a simulation model by incorporating likelihood functions to fit the model to time series of aggregate fishery catch, survey biomass, size compositions from the fishery and survey, and length-specific predator diets. Assessment model performance evaluation is in progress by comparing “true” biomass, abundance, and fishing mortality rates to model-estimated quantities using skill assessment metrics (e.g., correlation, RMSE, Modeling efficiency; Stow et al. 2009). Finally, the length-based multispecies model is being fit to actual observations from Georges Bank to evaluate operational potential within a particular region.

Methodology

The reviewed model is based on the simulation model Hydra (Gaichas et al. 2017), implemented in ADMB (Fournier et al. 2012). Hydra’s structure is derived from the length-based multispecies simulation model, LeMANS (Hall et al. 2006; Rochet et al. 2011), with additional options for growth, and recruitment functional forms and more detailed fishing fleets.

A key feature of Hydra that differentiates it from LeMANS and other length-based models is its treatment of length information. Attempting to balance model complexity and run time with available data and adequately modeling length-based processes, Hydra applies an equal number of length bins to all modeled species. The modeler specifies the overall number of bins (5 in the original simulation model), and the size in cm spanned by each bin for each species. The original Hydra simulation model used narrower length bins for smaller fish sizes and a length bin spanning a wider range of sizes for larger fish to efficiently represent changes size selectivity of predators and fisheries. The estimation model uses equal width bins equally dividing each species’ maximum observed length into the overall number of bins.

Modeler decisions on length bin number and width combined with individual species growth parameters both contribute to the model temporal scale, which internally calculated and set equal to the time required for

the fastest-growing species to grow out of its smallest length bin (if that is less than one year). Therefore, model timesteps will never exceed one year, but can be subannual.

For this review, the simulation model code was forked and an objective function added, with modifications to the data section to accommodate new data inputs and to the parameter section to estimate fishery and survey selectivity and catchability. As an estimation model, Hydra can be fitted to multiple data streams. Currently the objective function comprises the following components, catch (by species for each fleet), size composition of the catch (by species for each fleet), survey abundance index (by species for each survey), survey size composition (by species for each survey), survey diet composition (stomach weight by predator size bin for each predator species and survey), and recruitment (annual deviations from mean recruitment by species). A detailed description of the objective function equations is available [here](#).

Hydra input files were developed directly from `mskeyrun` datasets by modifying functions in the `hydradata` R package. The function `create_Rdata_mskeyrun.R` ([code](#)) allows the user to specify whether datasets should be constructed from Atlantis-simulated or real Georges Bank data, and the number of length bins to use for composition data, then creates an R data object. This data object is then used to create data and parameter input files using the function `hydradata::create_datpin_files()`.

Hydra output visualizations were developed to evaluate model fits with different settings and input datasets. The current diagnostics include model fits to each survey index, model fits to catch by fleet, comparisons of estimated and observed length compositions by year and in aggregate across years for each modeled species, residuals of fits to length compositions, and comparisons of estimated and observed diet composition. In addition, model outputs of estimated recruitment, total biomass, fishery and survey selectivity, fishing mortality, and predation mortality are produced. Initial model skill assessment compares model output total biomass with known total biomass for the simulated datasets.

All model code, data processing, and visualization is available online:

- Hydra model estimation version code https://github.com/thefaylab/hydra_sim
- Hydra estimation model data file generation <https://github.com/thefaylab/hydradata>
- Hydra model diagnostics https://github.com/thefaylab/hydra_diag

Data requirements

A common dataset for 10 Georges Bank species has been developed, as well as a simulated dataset for model performance testing. This `mskeyrun` data package holds both datasets. All modeling teams used these datasets. Group decisions on data are also documented below.

Years: 1968-2019

Area: Georges Bank (see map above)

Species: Atlantic cod (*Gadus morhua*), Atlantic herring (*Clupea harengus*), Atlantic mackerel (*Scomber scombrus*), Goosefish (*Lophius americanus*), Haddock (*Melanogrammus aeglefinus*), Silver hake (*Merluccius bilinearis*), Spiny dogfish (*Squalus acanthias*), Winter flounder (*Pseudopleuronectes americanus*), Winter skate (*Leucoraja ocellata*), and Yellowtail flounder (*Limanda ferruginea*)

Model description

Much of the material below is edited from from Gaichas et al. (2017), supplementary material:

Hydra is a length-based multispecies, multifleet model designed to provide simulated data for performance testing of simpler (non-size structured) multispecies assessment models and management procedures for the Northeast U.S. Continental Shelf.

We focus on body size rather than age because:

- Predation is a length-based process

- Harvesting is a length-based process
- Routine age determinations are not available for all ecologically and economically important species

Population dynamics For an individual species i , the number of individuals N in size class j at time $t + 1$ represent the survivors from that cohort from the previous time step (t) that have not grown into a larger size class, plus the number of individuals growing into size class j from smaller size classes during the time interval. We only consider transitions among immediately adjacent size classes. For size classes larger than the recruit (smallest) size class, the model for each species i can be written:

$$N_{i,j,t+1} = \sum_{j'} \phi_{i,j' \rightarrow j,t} S_{i,j',t} N_{i,j',t}$$

The proportion surviving from size class $j - 1$ to size class j for a species at time t is denoted $S_{j',t}$, $\phi_{j' \rightarrow j}$ is the probability of moving from size class $j - 1$ to size class j .

Growth and Time-in-Stage The values for $\phi_{j' \rightarrow j,t}$, the time required for an individual to grow between length bins are determined from the growth function. Growth can be assumed to follow a von Bertalanffy (see Hall et al. (2006)) or power function, either of which can be applied to any species. As in Hall et al. (2006), the model timestep is set up to equal the amount of time it takes for the fastest growing species/length bin combination to grow into the next length bin.

One of the simplest possible models for indeterminate growth with environmental effects is exponential:

$$l_{i,a,t} = \Psi_i a_i^{K_i}$$

where $l_{i,a,t}$ is the length of species i , age a and time t ; Ψ_i and K_i are growth model parameters.

The time required to grow through a specified length interval lb is:

$$\Delta t_i = \left[\frac{\Delta lb_i}{\Psi_i} \right]^{1/K_i}$$

and the expression for the probability ϕ of moving through the length interval is then:

$$\phi_i = \frac{1}{\Delta t_i}$$

Von Bertalanffy growth is modeled identically as in Hall et al. (2006):

$$l_{i,t} = L_{\infty i} (1 - e^{-vonBK_i(t-t_{0,i})})$$

with the time required to grow through a specified length interval:

$$\Delta t_i = \frac{1}{vonBK_i} \log \left[\frac{L_{\infty i} - lb_{lower,i}}{L_{\infty i} - lb_{upper,i}} \right]$$

and an identical expression for the probability of moving through the length interval as above.

Recruitment Average recruitment with annual deviations

In the most basic formulation, no spawner-recruit model is specified, but annual log deviations $\Lambda_{i,t}$ from mean log recruitment $\bar{R}_i$ determine annual recruitment levels:

$$N_{i,1,t} = e^{\bar{R}_i + \Lambda_{i,t}}$$

Mortality The survival rate is:

$$S_{i,j,t} = e^{-(M1_{i,j} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t})}$$

The rate of natural mortality is partitioned into two components: sources of natural mortality due to all unmodeled factors ($M1$) and predation mortality from species included in the model ($M2$).

The predation process is decomposed into size selectivity and species vulnerability in each area. The ρ , of prey species m size n for a given predator species i size j is:

$$\rho_{i,j,m,n} = \vartheta_{n,j} \lambda_{m,i}$$

where $\vartheta_{n,j}$ is the preference for a prey item of size n by predator size j , and $\lambda_{m,i}$ is the vulnerability of prey species m to predator species i . The parameter λ is set to either 0 or 1 depending on whether predator i is known to prey on species m .

The size preference function is:

$$\vartheta_{n,j} = \frac{1}{(w_n/w_j)\sigma_j\sqrt{2\pi}} e^{-\frac{[\log_e(w_n/w_j) - \Psi_j]^2}{2\sigma_j^2}}$$

where w_n is the weight at the midpoint of the length bin for a prey of size n , w_j is the weight at the midpoint of the length bin for a predator of size j , Ψ_j is the ‘preferred’ predator/prey weight ratio on a logarithmic scale, and σ_j^2 is the variance in predator size preference.

To estimate the daily food intake I for each predator, we use:

$$I_{i,j,t} = 24[\delta_i e^{\omega_i T}] \bar{C}_{i,j,k,t}$$

where the term in brackets gives the temperature (T)-dependent hourly consumption rate with parameters δ_i and ω_i and $\bar{C}_{i,j,k,t}$ is the mean stomach content weight (g) over a diel cycle.

The predation mortality rate on a particular prey species depends on the total consumption by predators, the suitability of that prey to each predator, and the total suitable prey biomass available to each of its predators. If the predation functional feeding response is assumed to be a Holling Type II, the $M2$ component on prey species m size n is then (Hall et al., 2006; Magnússon, 1995):

$$M2_{m,n,t} = \sum_i \sum_j I_{i,j,t} N_{i,j,t} \frac{\rho_{i,j,m,n}}{\sum_a \sum_b \rho_{i,j,a,b} W_{a,b} N_{a,b} + \Omega}$$

where a and b represent all prey species and sizes for predator i , W is the mean weight of prey a in size class b , Ω is ‘other’ food not explicitly included in the model, and all other terms are as above.

Fishing in the hydra estimation model has been re-implemented differently from the original simulation model. Similar to the original simulation model, multispecies fleets are implemented. However, rather than driving a simulation with fleet effort, we estimate fishing mortality rates (F) by fleet where there are one or more target species/stocks per fleet, with one “primary” stock identified for each fleet.

Annual fishing mortality rates (F) over time for the primary stock (modeled via average $Fbar$ and annual deviations $Fdev$) are specified as:

$$F_{f,i,y} = Fbar_{f,i} e^{(Fdev_{f,y})}$$

fleet-specific F for target stocks other than the primary stock modeled as offset from the primary stock via catchability q :

$$F_{f,j,y} = q_{f,j} \cdot F_{f,i,y}$$

We note this implies same fishing pattern across stocks within a fleet, which is a simplification that may need later adjustment. This could be modified with a multivariate random walk.

Logistic selectivity at length for each fleet is applied to all species caught by the fleet:

$$F_{f,i,t,l} = sel_{f,i,L(l)}$$

The selectivity for a specified gear type q is given by:

$$s_{i,j,q} = [1 + e^{-(c_{i,q} + d_{i,q}l)}]^{-1}$$

where $c_{i,j,q}$ and $d_{i,j,q}$ are model parameters and l is the midpoint of the j th length interval.

The yield from species i at time t is then:

$$Y_{i,t} = \sum_j \sum_q \frac{s_{i,j,q} F_{i,j,q,t}}{M1_{i,j} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t}} [1 - e^{-(M1_{i,j} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t})}] N_{i,j,t} W_{i,j}$$

where $W_{i,j,k}$ is the mean weight of an individual of species i and length j .

Objective Function

The model can be fitted to multiple data streams. Currently the objective function comprises the following components, catch (by species for each fleet), size composition of the catch (by species for each fleet), survey abundance index (by species for each survey), survey size composition (by species for each survey), survey diet composition (stomach weight by predator size bin for each predator species and survey), and recruitment (annual deviations from mean recruitment by species).

Catch Expected catch for species i in fleet q is summed over time steps to the fishing/calendar year y .

$$\hat{C}_{y,i,q} = \sum_{t \in y} \sum_j \frac{s_{i,j,q} F_{i,j,q,t}}{M1_{i,j,t} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t}} [1 - e^{-(M1_{i,j,t} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t})}] N_{i,j,t} W_{i,j,t}$$

$$\ln(C_{y,i,q}) \sim \mathcal{N}(\ln(\hat{C}_{y,i,q}), \sigma_{q,i,y}^2)$$

Catch length composition

$$\hat{p}_{q,i,y,j} = p_{q,i,y,j}^* / \sum_{j'} p_{q,i,y,j'}^*$$

$$p_{q,i,y,j}^* = \sum_{t \in y} \frac{s_{i,j,q} F_{i,j,q,t}}{M1_{i,j,t} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t}} [1 - e^{-(M1_{i,j,t} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t})}] N_{i,j,t}$$

Proportions at length (in numbers) in catch for each species in each fleet are modeled as multinomial, given an input annual sample size for each composition.

Survey abundance indices The predicted values for survey abundance rely on the timing of the survey within a year, catchability of the species in the survey, and the (length-based) survey selectivity. Survey g predicted values (in weight) are:

$$\hat{I}_{g,i,t} = q_{i,g,t} \sum_j j s_{i,j,g} N_{i,j,t} l_j W_{i,j,t}$$

The observed annual index values for each species are assumed to be lognormally distributed around the predicted values:

$$\ln(I_{g,i,t}) \sim \mathcal{N}(\ln(\hat{I}_{g,i,t}), \sigma_{g,i,t}^2)$$

Survey length composition

$$\hat{p}_{g,i,t,j} = p_{g,i,t,j}^* / \sum_{j'} p_{g,i,t,j'}^*$$

$$p_{g,i,t,j}^* = s_{i,j,g,t} N_{i,j,t}$$

Proportions at length (in numbers) in the survey for each species are modeled as multinomial, with an additional input sample size for each composition.

Prey proportions in survey diet Proportions by weight in stomach of predator i of size j of prey m :

$$\hat{\phi}_{g,t,i,j,m} = \phi_{g,t,i,j,m}^* / \left(\phi_{g,t,i,j,other}^* + \sum_m \phi_{g,t,i,j,m}^* \right)$$

$$\phi_{g,t,i,j,m}^* = \sum_n \rho_{i,j,m,n} N_{m,n,t} W_{m,n,t}$$

Proportions by prey species (in weight) in the survey are modeled as multinomial with an input sample size for each composition (potentially these data could be modeled as Dirichlet or delta-dirichlet, or...)

Penalties

Recruitment deviations A penalty to the objective function is added for the annual deviations from the stock-recruitment relationship for each species, which are assumed to be log-normal around the expected recruitment (equation 8):

$$\Lambda_{i,t} \sim \mathcal{N}(0, \tau_{\Lambda_i}^2)$$

where $\tau_{\Lambda_i}^2$ is the variance of the recruitment deviations for species i .

Estimated (or estimable) parameters

Parameter types and numbers of parameters by type (Nsp = number of species, Nlen = number of length bins, Nyr = number of years, Nfleet = number of fleets, Nsurvey = number of surveys) include:

1. Initial year numbers at size for each stock (Nsp * Nlen)
2. Average annual recruits for each stock (Nsp)
3. Recruitment deviations for each stock (Nsp * (Nyr-1))

4. Annual F for each fleet ($N_{\text{fleet}} * N_{\text{yr}}$)
5. Fishery catchability for non-primary stocks ($N_{\text{sp}} - N_{\text{fleet}}$)
6. Fishery logistic selectivity at length for each fleet ($2 * N_{\text{fleet}}$)
7. Survey catchability for each stock for each survey ($N_{\text{survey}} * N_{\text{sp}}$)
8. Survey logistic selectivity at length for each survey ($2 * N_{\text{survey}}$)

Not currently estimating (but could add):

1. M1 by stock (constant over age) (N_{stock})
2. M2 related parameters (e.g. vulnerability)

Diagnostics

Functions to visualize both model inputs and outputs are in development in the [hydra_diag github repository](#). At present, workflows for visualizing an individual model output and for comparing across model outputs are available, along with initial options to add model skill assessment (comparison with known simulated data).

Individual model diagnostics include plots of input survey biomass, catch, and length data (not aggregated by length bin). Model fits to aggregate survey indices and catch by fleet, fits to annual and aggregate (all years) survey and fishery length compositions (length bins), and diet composition for predator and length bin are visualized. In addition, effective sample size is compared with input sample size for length and diet compositions, and both Pearson and One Step Ahead (OSA, Trijoulet et al. (2023)) residuals are calculated for fits to all length compositions. Model derived quantities (estimated recruitment, biomass, fishing mortality, predation mortality, and survey and fishery selectivity) are also plotted.

Modeling approach

Given the current development level of Hydra as an estimation model, current objectives are to demonstrate that the model runs and generates the desired output. Priorities for further development should ideally be identified during the ECES WGSAM review. Hydra is being fit to both simulated and real Georges Bank datasets to develop and test estimation functions.

For simulated data our initial species selection includes 11 single species groups from the Norwegian Barents Sea (NOBA) Atlantis model (Hansen et al. 2016, 2019). These groups are fully age structured. All but two of them are fished. The full process for generating the simulated dataset is described on the `mskeyrun` R package documentation [at this link](#). Hydra was fit to an 80 years of simulated survey and catch index, size, and diet composition data for these 11 simulated stocks using 2 fleets (cod and other), with both 5 and 10 length bins for each stock. The simulated datasets were used to test incrementally added estimation functionality, and will be used for model skill assessment.

For Georges Bank, the model was fit to 10 stocks using 2 fleets (groundfish and pelagic), with 5 length bins for each stock. Time series for fitting included 1978 - 2019. The model was fit to total catch (summed retained + discard), catch size compositions, annual Fall and Spring NEFSC bottom trawl survey abundance indices (pre-processed for vessel differences via calibration), survey size compositions, survey and stomach content diet composition (proportion by weight). M1 and M2-related mortality parameters are fixed inputs (for now).

Initial fits to real Georges Bank data have been for testing only, but several examples were presented during the review. Please see the “Hydra Results” document for an overview.

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